
Conestoga College
MATH73235: Advanced Topics in Mathematics
Spring 2025 Midterm 2
July 15, 2025

Exam Duration: 5pm - 7pm.

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This exam contains 10 pages (including this cover page) and 5 questions. The total number of possible points is 50. Enter your answers in the space provided.

- **Organize your work**, in a reasonably neat and coherent way, in the space provided.
- **Mysterious or unsupported answers will not receive full credit.** Correct answers should be supported by calculations, explanations, or algebraic work to receive full credit; incorrect answers supported by substantially correct calculations and explanations may still receive partial credit.
- **Provide exact answers** unless otherwise instructed.
- **Simplify all answers as much as possible.** This means that you need to combine like terms, reduce fractions, etc.
- **Be sure to state units** for applied problems.
- **Clearly identify your answer** for each problem.

Question	Points	Score
1	10	3
2	10	10
3	10	10
4	10	9
5	10	8
Total:	50	40

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1. (10 points) **Conceptual Questions.** Write an equation, an expression, or a short sentence to describe each of these concepts.

- (a) The expression for the volume integral $\iiint_T f(x, y, z) dx dy dz$ when the coordinates are changed from (x, y, z) to some other coordinates (u, v, w) .

Solution.

✓
2 $\iiint_T f(x, y, z) = \iiint f(x(u, v, w), y(u, v, w), z(u, v, w)) \cdot \frac{\partial x \partial y \partial z}{\partial u \partial v \partial w} du dv dw$

The notation is: $\frac{\partial(x, y, z)}{\partial(u, v, w)}$, but I will accept $\square$

- (b) The main equation for **Stokes' Theorem** for a vector field $\vec{F}$ in a surface S with a boundary curve C traversed such that S remains on the left of C .

Solution.

0 $\oiint_S \vec{F} \cdot \vec{n} dA = \iiint_V \vec{\nabla} \cdot \vec{F} dV$ X This is Divergence Theorem

Stokes is $\oiint_S (\vec{\nabla} \times \vec{F}) \cdot \vec{n} dA = \oint_C \vec{F} \cdot d\vec{r}$ $\square$

- (c) An expression for a vector $\vec{N}$ that is normal to the surface $z = x^2 + y^2 = 0$.

Solution.

0 $\vec{N} = \vec{\nabla} \times \vec{F}$ X $\begin{bmatrix} \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ F_1 & F_2 & F_3 \end{bmatrix} \Rightarrow \hat{n} = \frac{\vec{N}}{|\vec{N}|}$ either parameterize or use $\vec{\nabla} g$ $\square$

$f(x, y, z) = x^2 + y^2 + z$
 $[x, y, x^2 + y^2]$

- (d) The vector field $\vec{F}$ polar coordinates written as $F_r \hat{e}_r + F_\theta \hat{e}_\theta$ if in Cartesian coordinates we have $\vec{F} = F_1 \hat{i} + F_2 \hat{j} = (x^2 + y^2) \hat{i} - 2y \hat{j}$. Hint: $F_r = F_1 \cos \theta + F_2 \sin \theta$, $F_\theta = -F_1 \sin \theta + F_2 \cos \theta$.

Solution.

1 $F_r = (x^2 + y^2) \cos \theta - 2y \sin \theta$, $F_\theta = -(x^2 + y^2) \sin \theta - 2y \cos \theta$

$\hookrightarrow (x^2 + y^2) \cos \theta$ use $x^2 + y^2 = r^2$, and $y = r \cos \theta$ $\square$

- (e) The expression for the divergence $\nabla \cdot \vec{F}$ in spherical coordinates (ρ, θ, ϕ) if we are given that the divergence of the same vector field is $\nabla \cdot \vec{F} = r^2 + z^2$ in cylindrical coordinates (r, θ, ϕ) .

Solution.

0 $\nabla \cdot \vec{F} = \frac{1}{(\rho \sin \theta) \rho} \left[\frac{\partial}{\partial \rho} (\rho \sin \theta \rho) (F_\rho) \right]$ X

$\hookrightarrow \nabla \cdot \vec{F} = \frac{1}{\rho^2 \sin \theta} \left[\frac{\partial}{\partial \rho} (\rho \sin \theta) (\rho^2) \right]$

$r^2 = x^2 + y^2$ $\vec{\nabla} \cdot \vec{F} = r^2 + z^2 = \rho^2$
 $\hookrightarrow x^2 + y^2 + z^2$
in spherical $= \rho^2 = x^2 + y^2 + z^2$ $\square$

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2. (10 points) Use the appropriate Lamé coefficients to compute the Laplacian of the scalar function

$$f(x, y, z) = x^2 + y^2 + z^2$$

in both Cylindrical and Spherical coordinates. You must compute all the required partial derivatives.

Solution.

(Cylindrical) $h_r = 1, h_\theta = r, h_z = 1$

$$\nabla_{fz}^2 \frac{1}{(1)(r)(1)} \left[\frac{\partial}{\partial r} \left(\frac{(r)(1)}{(1)} \right) \left(\frac{\partial}{\partial r} (r^2) \right) + \frac{\partial}{\partial \theta} \left(\frac{(1)(1)}{(r)} \right) \cdot \frac{\partial}{\partial \theta} (0) + \frac{\partial}{\partial z} \left(\frac{(1)(r)}{(1)} \right) \cdot \frac{\partial}{\partial z} (z^2) \right]$$

$$f(x, y, z) = x^2 + y^2 + z^2$$

$$\hookrightarrow x^2 + y^2 = r^2$$

$$\hookrightarrow f(x, y, z) = r^2 + z^2$$

therefore $\frac{\partial f}{\partial \theta} = 0$

$$\hookrightarrow \nabla_{fz}^2 \frac{1}{r} \left[\frac{\partial}{\partial r} (r(2r)) + \frac{\partial}{\partial z} (r \cdot 2z) \right]$$

$$\hookrightarrow \nabla_{fz}^2 \frac{1}{r} \left[\frac{\partial}{\partial r} (2r^2) + \frac{\partial}{\partial z} (2rz) \right]$$

$$\hookrightarrow \nabla_{fz}^2 \frac{1}{r} (4r + 2r) \Rightarrow \nabla_{fz}^2 \frac{1}{r} (6r) \Rightarrow \boxed{\nabla_{fz}^2 6} \checkmark$$

Continued on pg 9.

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3. (10 points) Use **Spherical coordinates** to find the surface area of the cone $\phi = \frac{\pi}{6}$ with $0 \leq \rho \leq 2$.
Solution.

there fore we use

$$\phi = \text{constant} = \frac{\pi}{6}$$

$$dA = \rho^2 \sin \phi \, d\rho \, d\theta \Rightarrow dA = (1)(\rho \sin \phi) \, d\rho \, d\theta$$

$$\hookrightarrow dA = \rho \sin\left(\frac{\pi}{6}\right) \, d\rho \, d\theta$$

$$0 \leq \rho \leq 2$$

$$0 \leq \theta < 2\pi$$

$$\hookrightarrow dA = (\rho) \left(\frac{1}{2}\right) \, d\rho \, d\theta \Rightarrow dA = \frac{1}{2} \rho \, d\rho \, d\theta$$

$$\Rightarrow \int_{\theta=0}^{2\pi} \int_{\rho=0}^2 \frac{1}{2} \rho \, d\rho \, d\theta \Rightarrow \frac{1}{2} \int_0^{2\pi} d\theta \int_0^2 \rho \, d\rho$$

$$\hookrightarrow \frac{1}{2} (2\pi) \left[\frac{\rho^2}{2} \right]_0^2 \Rightarrow \frac{1}{2} (2\pi) \left(\frac{4}{2} - \frac{0}{2} \right)$$

$$\hookrightarrow \frac{1}{2} (2\pi) (2) = 2\pi \checkmark$$

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4. (10 points) Verify Gauss' Divergence Theorem by computing the following surface integral:

$$\iint_S \vec{F} \cdot \hat{n} \, dA,$$

and the volume integral:

$$\iiint_T (\nabla \cdot \vec{F}) \, dV,$$

where $\vec{F} = x\hat{i} + y\hat{j} + z\hat{k}$, S is the surface of a sphere of radius 2 that forms the boundary of T .

Hint: This problem may be easier to solve in the spherical coordinates. Note that in spherical coordinates, $\vec{F} = \rho\hat{e}_\rho$, while for S , $\hat{n} = \hat{e}_\rho$, and thus

$$\vec{F} \cdot \hat{n} = (\rho\hat{e}_\rho) \cdot (\hat{e}_\rho) = \rho.$$

Solution.

First we get the divergence of the vector field

$$\hookrightarrow \nabla \cdot \vec{F} = \frac{\partial}{\partial x} f + \frac{\partial}{\partial y} f + \frac{\partial}{\partial z} f = \frac{\partial}{\partial x} (x) + \frac{\partial}{\partial y} (y) + \frac{\partial}{\partial z} (z)$$

$$\hookrightarrow \nabla \cdot \vec{F} = (1) + (1) + (1) = 3 \quad \text{now sub back in } \iiint 3 \, dV$$

$$\hookrightarrow dV = \frac{4}{3} \pi r^3 \Rightarrow \frac{4}{3} \pi (2^3) = \frac{4}{3} \cdot 8\pi = \frac{32}{3} \pi \quad r=2$$

(Note: The handwritten solution includes a correction: "This is not dV, dV = \int^2 \sin\theta \, d\theta \, d\phi \, dr". The circled term \frac{4}{3}\pi r^3 is crossed out and replaced with the correct volume calculation.)

$$\hookrightarrow \iiint_V 3 \left(\frac{32}{3} \pi \right) \Rightarrow \boxed{32\pi \checkmark}$$

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Continue problem 4

Solution.

Surface integral $\iint_S \vec{F} \cdot \hat{n} \, dA$ $\vec{F} \cdot \hat{n} = \rho$

$dA = h_\theta h_\phi \, d\theta d\phi \Rightarrow dA = \rho \sin \theta \cdot \rho \, d\theta d\phi$ $r=2$

$\hookrightarrow dA = \rho^2 \sin \theta \, d\theta d\phi \Rightarrow dA = 4 \sin \theta \, d\theta d\phi$ ✓

$4 \int_0^{2\pi} d\theta \int_0^\pi \sin \theta \, d\theta \Rightarrow 4(2\pi) [-\cos \theta]_0^\pi$ ✓

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5. (10 points) In the first quadrant of the Cartesian plane (i.e., $x \geq 0, y \geq 0$), we may define a set of coordinates known as the hyperbolic coordinates (u, v) , with transformation function

$$x = ve^u, \quad y = ve^{-u}.$$

Find the area of the triangular region R given by

$$0 \leq v \leq u, \quad 0 \leq u \leq 3.$$

Solution.

$$x(u, v) = ve^u$$

$$y(u, v) = ve^{-u}$$

$$\frac{d(x, y)}{d(u, v)} = \left| \det \begin{bmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{bmatrix} \right|$$

$$f_c(x, y)$$

$$f_c(ve^u, ve^{-u})$$

$$\Rightarrow \begin{vmatrix} \frac{d}{du}(ve^u) & \frac{d}{dv}(ve^u) \\ \frac{d}{du}(ve^{-u}) & \frac{d}{dv}(ve^{-u}) \end{vmatrix} = \begin{vmatrix} ve^u & e^u \\ ve^{-u} & -e^{-u} \end{vmatrix} = (e^{-u})(ve^u) - (e^u)(ve^{-u})$$

missed

$$\Rightarrow (ve^u e^{-u} - ve^u e^{-u}) = 1 \quad \times$$

$$\iint_S f(x, y) \Rightarrow \iint_R f(x(u, v), y(u, v)) \cdot \det \, du \, dv$$

$$\Rightarrow \int_{v=0}^u \int_{u=0}^3 (ve^u)(ve^{-u})(1) \, du \, dv \Rightarrow \int_{u=0}^3 \int_{v=0}^u v^2 \cdot e^u \cdot e^{-u} \, dv \, du$$

$$\Rightarrow \int_0^3 \left[\frac{v^3}{3} \cdot e^u \cdot e^{-u} \right]_0^u \, du \Rightarrow \int_0^3 \frac{u^3}{3} \cdot e^u \cdot e^{-u} \, du \Rightarrow \left[\frac{u^3}{3} \cdot e^u \cdot e^{-u} \right]_0^3$$

$$\Rightarrow 27 \cdot e^3 \cdot e^{-3} = 27$$

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Question 2 Continued)

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Spherical coordinate)

$$h_{\rho} = 1, \quad h_{\theta} = \rho \sin \theta, \\ h_{\phi} = \rho$$

$$\hookrightarrow \nabla_{f_z}^2 = \frac{1}{(1)(\rho \sin \theta)(\rho)} \left[\frac{d}{d\rho} \left(\frac{\rho \cdot \rho \sin \theta}{(1)} \cdot \frac{d}{d\rho} (\rho^2) + 0 + 0 \right] \quad f(x, y, z) = \rho^2 \quad \checkmark$$

$$\hookrightarrow \rho^2 = x^2 + y^2 + z^2$$

$$\hookrightarrow \nabla_{f_z}^2 = \frac{1}{\rho^2 \sin \theta} \left[\frac{d}{d\rho} (\rho^2 \sin \theta (z\rho)) \right]$$

$$\hookrightarrow \text{therefore } \frac{\partial f}{\partial \theta} = \frac{d f}{d \theta} = 0$$

$$\hookrightarrow \nabla_{f_z}^2 = \frac{1}{\rho^2 \sin \theta} \left[\frac{d}{d\rho} (z\rho^3 \sin \theta) \right] \Rightarrow \frac{d}{d\rho} (z\rho^3 \sin \theta) = 6\rho^2 \sin \theta \quad \checkmark$$

$$\hookrightarrow \nabla_{f_z}^2 = \frac{1}{\cancel{\rho^2 \sin \theta}} (\cancel{6\rho^2 \sin \theta}) \Rightarrow \boxed{\nabla_{f_z}^2 6} \quad \checkmark$$

Formula Sheet

General Coordinates:

$$(x_1, x_2, x_3) \rightarrow (q_1, q_2, q_3):$$

$$x_1 = x_1(q_1, q_2, q_3), \quad x_2 = x_2(q_1, q_2, q_3), \quad x_3 = x_3(q_1, q_2, q_3)$$

$$\text{Derivative Matrix: } D = \begin{bmatrix} \frac{\partial x_1}{\partial q_1} & \frac{\partial x_1}{\partial q_2} & \frac{\partial x_1}{\partial q_3} \\ \frac{\partial x_2}{\partial q_1} & \frac{\partial x_2}{\partial q_2} & \frac{\partial x_2}{\partial q_3} \\ \frac{\partial x_3}{\partial q_1} & \frac{\partial x_3}{\partial q_2} & \frac{\partial x_3}{\partial q_3} \end{bmatrix}$$

$$\text{Lamé coefficients: } h_j^2 = \sum_{k=1}^3 \left(\frac{\partial x_k}{\partial q_j} \right)^2$$

$$\text{Jacobian: } \frac{\partial(x_1, x_2, x_3)}{\partial(q_1, q_2, q_3)} = |J| = |\det D| = |h_1 h_2 h_3|$$

$$\text{Line element: } ds^2 = h_1^2 dq_1^2 + h_2^2 dq_2^2 + h_3^2 dq_3^2$$

$$\text{Volume element: } dV = h_1 h_2 h_3 dq_1 dq_2 dq_3$$

$$\text{Area element for surface with } q_j = \text{const} \text{ (i, j, k distinct): } dA = h_i h_k dq_i dq_k$$

$$\text{Gradient for scalar function } f: \nabla f = \frac{1}{h_1} \frac{\partial f}{\partial q_1} \hat{e}_{q_1} + \frac{1}{h_2} \frac{\partial f}{\partial q_2} \hat{e}_{q_2} + \frac{1}{h_3} \frac{\partial f}{\partial q_3} \hat{e}_{q_3}$$

$$\text{Divergence and curl of vector field } \vec{F} = F_{q_1} \hat{e}_{q_1} + F_{q_2} \hat{e}_{q_2} + F_{q_3} \hat{e}_{q_3}:$$

$$\nabla \cdot \vec{F} = \frac{1}{h_1 h_2 h_3} \left[\frac{\partial}{\partial q_1} (h_2 h_3 F_{q_1}) + \frac{\partial}{\partial q_2} (h_3 h_1 F_{q_2}) + \frac{\partial}{\partial q_3} (h_1 h_2 F_{q_3}) \right]$$

$$\nabla \times \vec{F} = \frac{1}{h_1 h_2 h_3} \begin{vmatrix} h_1 \hat{e}_{q_1} & h_2 \hat{e}_{q_2} & h_3 \hat{e}_{q_3} \\ \frac{\partial}{\partial q_1} & \frac{\partial}{\partial q_2} & \frac{\partial}{\partial q_3} \\ h_1 F_{q_1} & h_2 F_{q_2} & h_3 F_{q_3} \end{vmatrix}$$

Laplacian of scalar field f :

$$\nabla^2 f = \frac{1}{h_1 h_2 h_3} \left[\frac{\partial}{\partial q_1} \left(\frac{h_2 h_3}{h_1} \frac{\partial f}{\partial q_1} \right) + \frac{\partial}{\partial q_2} \left(\frac{h_3 h_1}{h_2} \frac{\partial f}{\partial q_2} \right) + \frac{\partial}{\partial q_3} \left(\frac{h_1 h_2}{h_3} \frac{\partial f}{\partial q_3} \right) \right].$$

Cylindrical Coordinates:

$$x = r \cos \theta, \quad y = r \sin \theta, \quad z = z.$$

$$h_1 = h_r = 1, \quad h_2 = h_\theta = r, \quad h_3 = h_z = 1.$$

Spherical Coordinates:

$$x = \rho \cos \theta \sin \phi, \quad y = \rho \sin \theta \sin \phi, \quad z = \rho \cos \phi$$

$$h_1 = h_\rho = 1, \quad h_2 = h_\theta = \rho \sin \phi, \quad h_3 = h_\phi = \rho.$$